

JKBOSE 2026

CLASS 11TH

BIOLOGICAL CLASSIFICATION

LECTURE-07

BOTANY

ARSHID BHAT SIR



Topics To Covered

1 Kingdom Fungi ✓

KINGDOM FUNGI → "Mycota"

- Study of fungi — "mycology."
- It includes
 - achlorophyllous (without chlorophyll)
 - thalloid (Root, stem, leaf, not differentiated)
 - Spore producing
 - Eukaryotic heterotrophic organisms.

⊗ Cell wall : +nt

- Composed of chitin and polysaccharides.

Habitat: Cosmopolitan (everywhere)

- grow in warm and humid places.

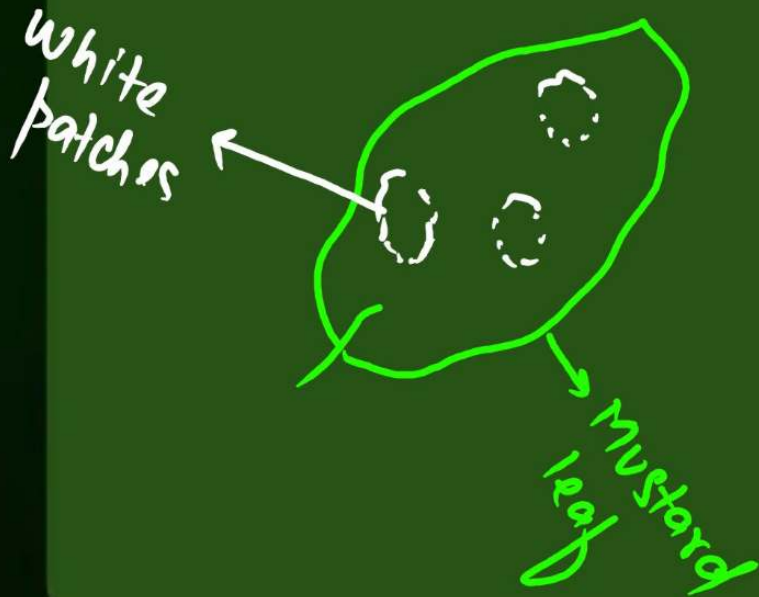
(*) Majority of fungi are Saprophytic — grows on moist bread
and
rotten fruits.

Ex Penicillium (Source of Antibiotic — penicillin)

(*) Some fungi are parasitic — grows on Plants
Animals
humans.

Ex. Albugo candidus

↓
Causes white patches on mustard leaves i.e, why it is AKA white rust



⊗ Some fungi are **poisonous** ex. "Toadstools"

⊗ Common **edible** mushroom — Agaricus bisporous (Button mushroom)

BODY FORMS



• Multicellular except. "yeast" (Saccharomyces cerevisiae)
↓
"Unicellular"

AKA Baker's yeast (∵ it is used in Bakery)
Brewer's yeast (∵ it is used in Brewing industry)

⊛

Hyphae - plural
Hypha - singular

→ Basic thread like str.
→ Slender
→ Two types:



a. Aseptate hypha: • NO Septa (wall) b/w Nuclei.
• Multinucleated (coenocytic)



b. Septate hypha: • Septa (wall) +nt b/w Nuclei.

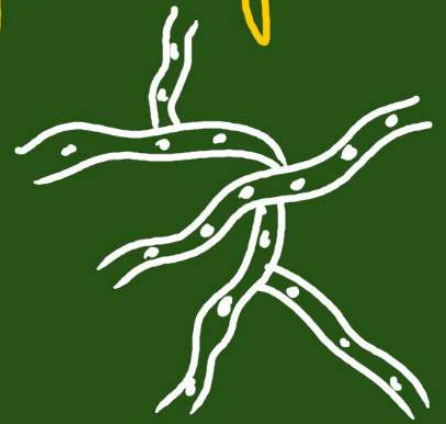


⊛ Many hyphae combine together to form a "Mycellium"

↓
Network of hyphae

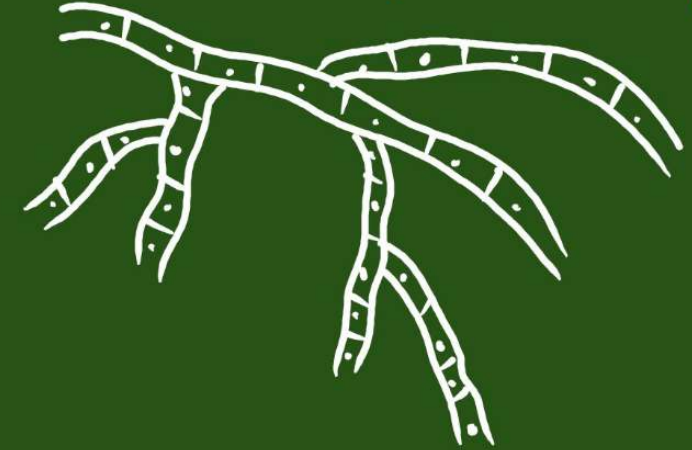
Mycellium

Aseptate mycellium



↓
Network of aseptate hyphae.

Septate mycellium



↓
Network of Septate hyphae.

NUTRITION

(Heterotrophic)



① Saprophytic (maximum)

② Parasitic

③ Symbiotic

LICHEN

→ Symbiotic association b/w algae and fungi.
→ Two partners are involved:

i) Algal partner (phycobiont) — perform photosynthesis

ii) Fungal partner (mycobiont) — absorbs H_2O and minerals from soil.

MYCORRHIZA

→ Symbiotic association b/w fungi and roots of higher plants (eg. Pinus).

→ Fungi — absorbs phosphorus from soil for plant.

→ Pinus — performs photosynthesis.

PIGMENTS

≡ No photosynthetic pigments.

≡ Colour providing pigments eg. Carotenoids are +nt.

Stored food:



Glycogen.

"PLZZ REMEMBER"

04 classes of Fungi

- ① Phycomycetes.
- ② Ascomycetes.
- ③ Basidiomycetes.
- ④ Deuteromycetes.

REPRODUCTION

≡ By 03 ways:

- a. vegetative reproduction.
- b. Asexual " "
- c. Sexual reproduction.

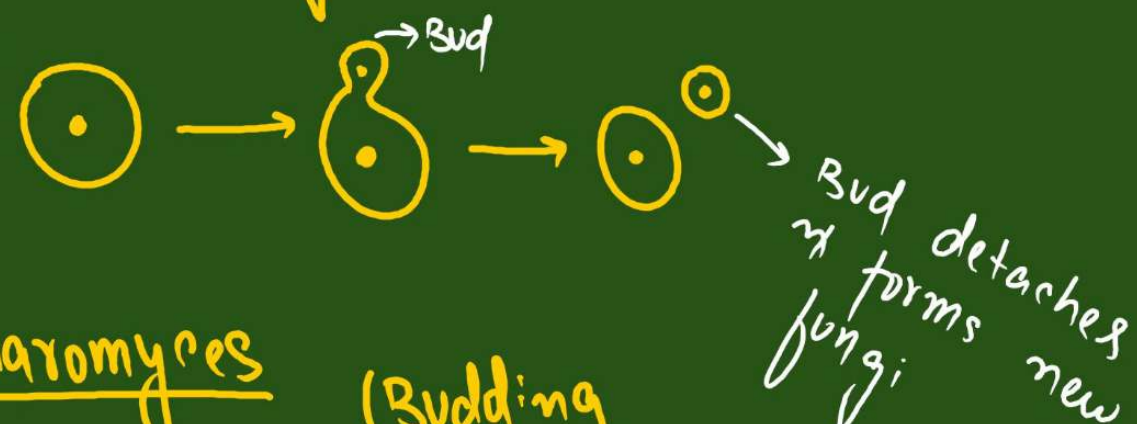
VEGETATIVE REPRODUCTION

= 3 types :

i) Fragmentation: ex. Filamentous forms of fungi.

ii) Fission: 

ex. Schizosaccharomyces (Fission yeast).

iii) Budding: 

ex. Saccharomyces (Budding yeast).

ASEXUAL REPRODUCTION

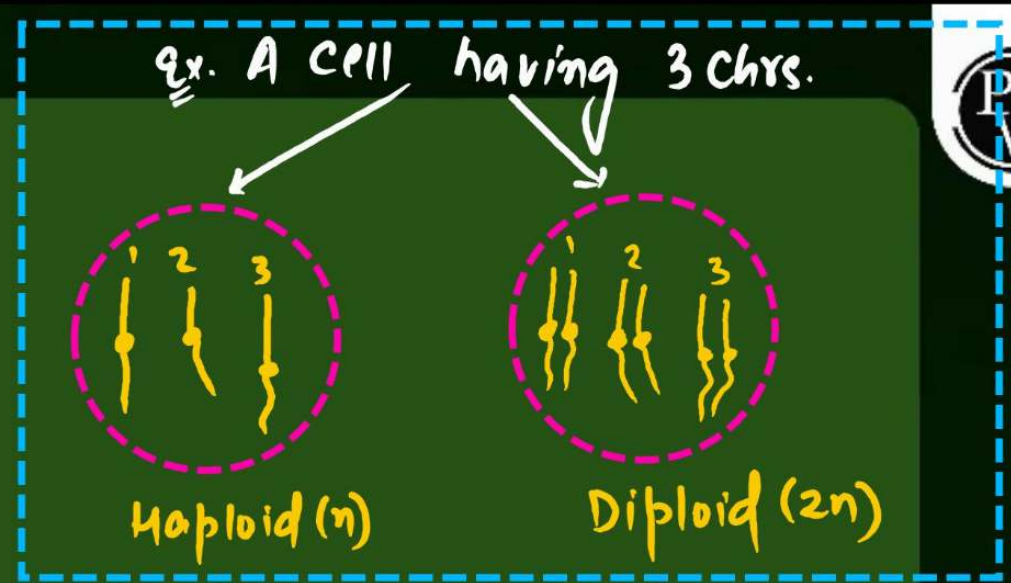
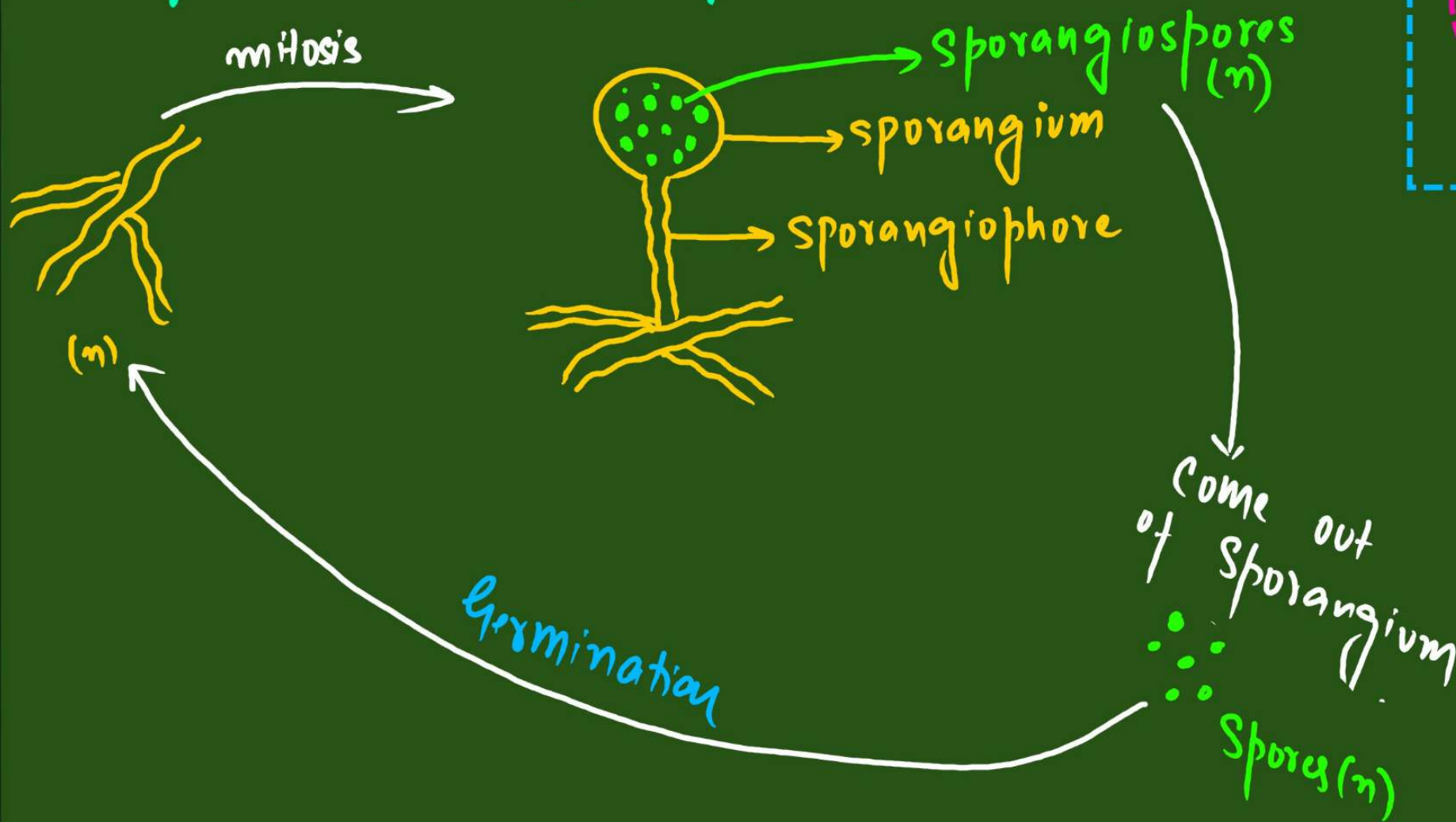
• By different types of asexual spores like :

a. Sporangiospores  Zoospores / Planospores
Aplanospores.

b. Conidia / conidiospores (Most common asexual spore in fungi).

a. Sporangiospores:-

- formed inside Sporangium. (Endogenous)



Sporangiospore — 02 types:

①. Zoospore / planospore :-

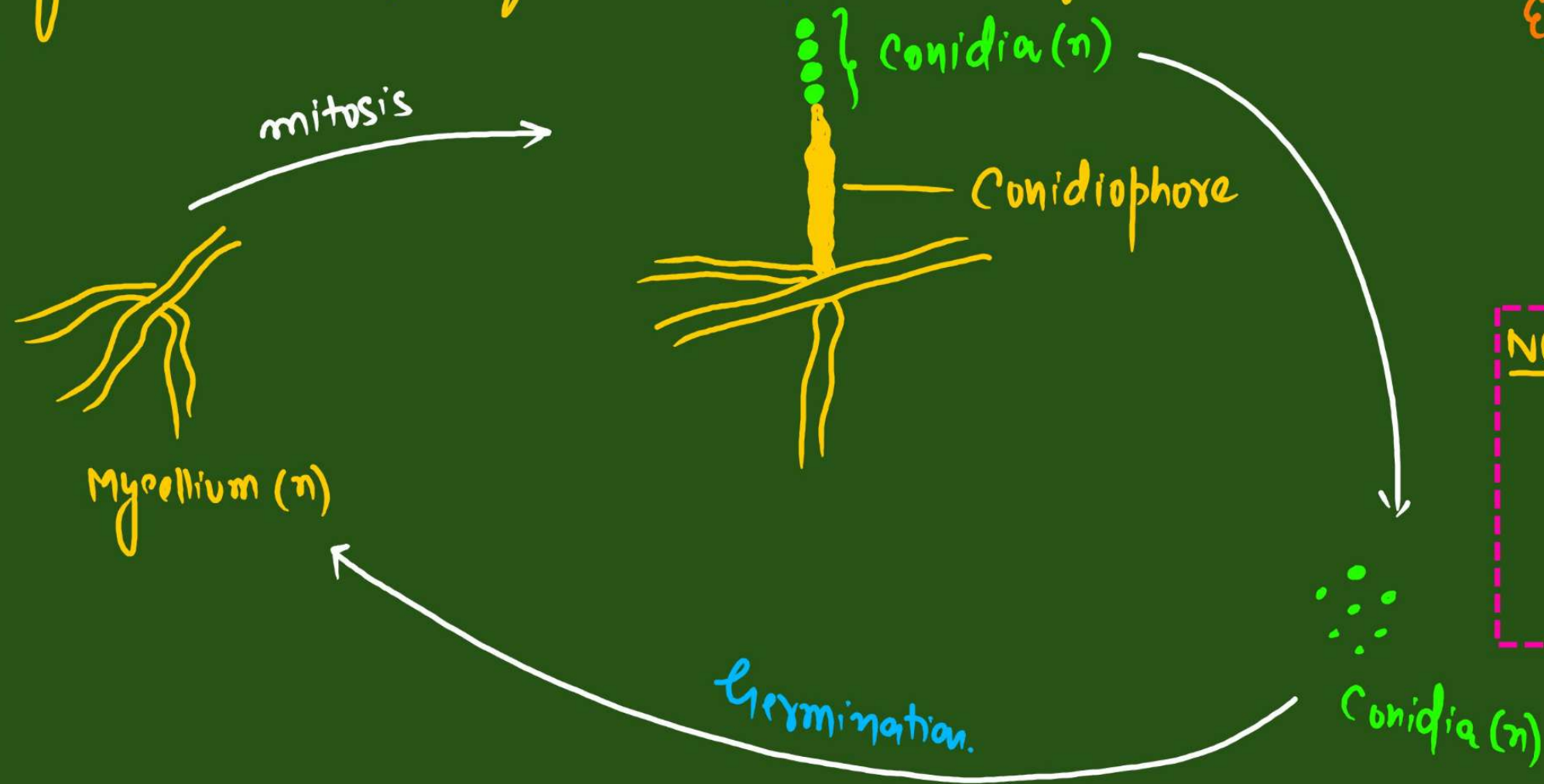
- Flagellum +nt (motile)
- Naked (No wall).
- Ex. Aquatic members of phycomycetes

②. Aplanospore :

- Flagellum -nt
- Non-motile
- wall +nt.
- Ex. Terrestrial members of phycomycetes (Ex Rhizopus, Mucor).

5. Conidia:

- formed at tip of conidiophore (exogenous)



Ex. **Ascomycetes**

Deuteromycetes.

Note:- In **Basidiomycetes**
Asexual spores are
generally Absent.

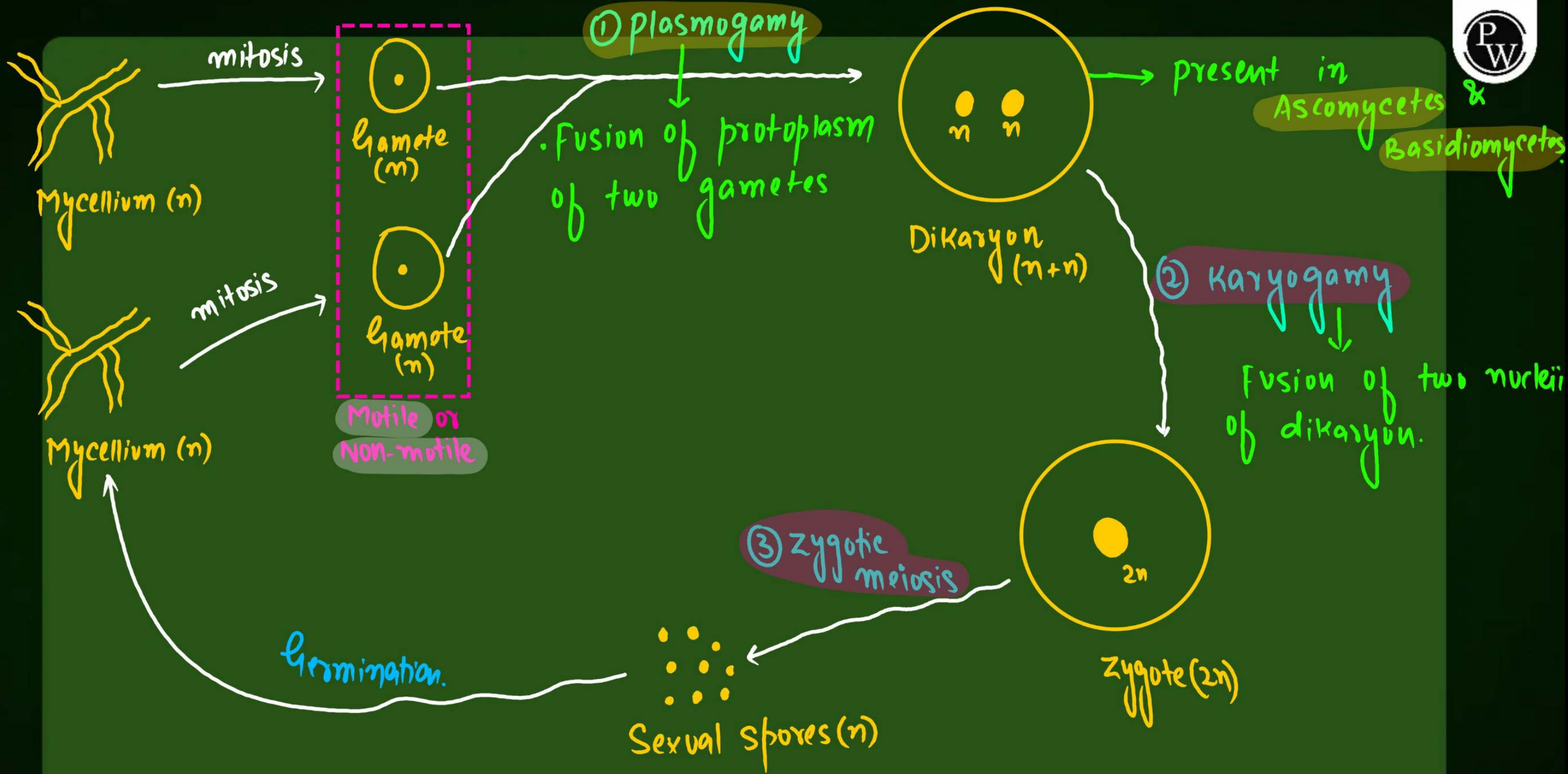
SEXUAL REPRODUCTION

≡ It involves 03 Stages:

a. Plasmogamy.

b. Karyogamy.

c. Zygotic meiosis.



QUESTIONS

Which one of the following is wrong for fungi?

[2016 II]

- 1 They are heterotrophic ✓
- 2 They are both unicellular and multicellular ✓
- 3 They are eukaryotic ✓
- ~~4~~ All fungi possess a purely cellulosic cell wall ✗

QUESTIONS

In most of the fungi, the food material is stored in the form of:

[2000]

- 1 Starch
- 2 Proteins
- ☒ 3 Glycogen
- 4 Chitin

QUESTIONS

Which of the following statements is correct?

[MR* 2021]

- 1 Fusion of protoplasts between two motile or non-motile gametes is called plasmogamy
- 2 Organisms that depend on living plants are called saprophytes
- 3 Some of organisms can fix atmospheric nitrogen in specialized cells called sheath cells.
- 4 Fusion of two cells is called Karyogamy.

Thank
You